



Govt. M.A.O. Graduate College, Lahore.

Re-MIDTERM EXAMS (January 2024)

Department/Discipline: Physics

Paper: Mechanics-I

Course code: PHY-111

Teacher's Name: Dr. Mureed Hussain

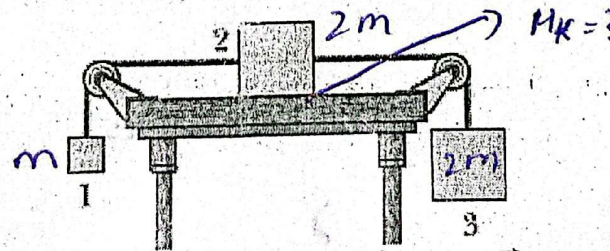
SEMESTER: 1st (MORNING)

Max. Marks: 25

Total Time: 1:00 Hour

Student's Name: ABOUL RAUF College Roll No. 24120 PU Roll No. _____

- 1) What is angular displacement? Is it a vector quantity or a scalar quantity? Justify your answer.
- 2) What is trajectory of a projectile? Derive an equation for the trajectory of a projectile?
- 3) When the three blocks in the figure below are released from rest, they accelerate with a magnitude of 0.300 m/s^2 . Block 1 has mass M , block 2 has mass $2M$ and block 3 has mass $2M$. What is the coefficient of kinetic friction between block two and the table?



- 4) In the product $\vec{F} = q \vec{v} \times \vec{B}$, take $q=2$, $\vec{v}=4.0\hat{i}+6.0\hat{j}+2.0\hat{k}$ and $\vec{F}=-20\hat{i}+12\hat{j}+4.0\hat{k}$. What then is $\vec{B}$ in unit vector notation if $B_y=B_z$?
- 5) A mass spring system is compressed. Calculate the work done in compressing the system and also find the potential energy stored in it?



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MIDTERM EXAMS (December 2024)

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Course code: PHY-111

Teacher's Name: Dr. Mureed Hussain

SEMESTER: 1st (MORNING)

Max. Marks: 25

Total Time: 1:00 Hour

Student's Name: ABDUL RAUF

College Roll No. 24120

PU Roll No. _____

Short Questions

(5 x 3 = 15)

- 6) A cube is a three dimensional geometrical object that has same height, width and length. Use a proper scale to write down the position vectors of all the eight corners of a cube with side length L ?
- 7) A key is dropped from a bridge that is 10 m high above the surface of water. The key drops into a model boat which is moving with a constant velocity. When the key is dropped, the boat was 10 m away from the point of impact. What is the velocity of the boat?
- 8) What is a force? Explain how Newton's laws explain the motion of an object under forces?
- 9) What is work? How is it related to the energy of a system?
- 10) What is potential energy? Explain how the potential energy is defined for a system acted upon by conservative forces only? Give examples.

Long Questions

(2 x 5 = 10)

- 1) What is uniform circular motion? Show that for an object in circular motion $a_c = \frac{v^2}{r}$? (5)
- 2) What is center of mass? Calculate the center of mass of a uniform cylinder? (5)